Macroeconomía Dinámica

SIR Models

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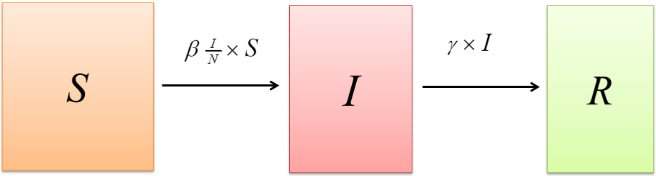
Basic SIR Model

- In every period, each individual is in one of three states:

$$\begin{aligned} S &= \text{susceptible} \\ I &= \text{infectious} \\ R &= \text{removed (recovered or dead)} \end{aligned}$$

- Graphically:



- Notice that the removed cannot become susceptible or infectious again (permanent immunity).
- Remark: infectious $\neq$ infected; an infected person may not transmit the disease to others.

Basic SIR Model

- Constant population N (including the deceased):

$$S_t + I_t + R_t = N$$

- Initial conditions:

$$S_0 = (1-\epsilon)N \cong N, \quad I_0 = \epsilon N > 0, \quad R_0 = 0$$

- Laws of motion (discrete-time version):

$$\begin{aligned} S_{t+1} - S_t &= -\beta \left(\frac{I_t}{N} \right) S_t \\ I_{t+1} - I_t &= \beta \left(\frac{I_t}{N} \right) S_t - \gamma I_t \\ R_{t+1} - R_t &= \gamma I_t \end{aligned}$$

  - Notice that these equations are consistent with a constant N.
- Time is usually measured in days.

Basic SIR Model

- Interpretation of $\beta\left(\frac{I_t}{N}\right)S_t$.
  - β is the average number of contacts (sufficient for transmission), per unit time (per day), of one individual.
  - $\beta\left(\frac{I_t}{N}\right)$ is the average number of contacts with infectious people, per unit time (per day), of one susceptible person.
  - $\beta\left(\frac{I_t}{N}\right)S_t$ is the number of new infectious cases, per unit time (per day), due to S_t susceptibles.
- Supose $\beta = \frac{1}{5} = 0.2$, $\frac{I_t}{N} = \frac{10}{40} = \frac{1}{4} = 0.25$ and $S_t = 20$ million. Then:
  - $\beta = \frac{1}{5}$ means that, on average, there's one contact every five days.
  - $\beta\left(\frac{I_t}{N}\right) = \frac{1}{5}\frac{1}{4} = \frac{1}{20}$ means that, on average, there's one contact with an infectious person every twenty days.
  - $\beta\left(\frac{I_t}{N}\right)S_t = \frac{1}{20}20 = 1$ means that, on average, one million people become infectious each day (and stop being susceptible).

Basic SIR Model

- Alternatively:
  - $\beta\left(\frac{S_t}{N}\right)$ is the average number of susceptibles that get the disease (and become infectious) from one infectious individual, per unit time (per day).
  - $\beta\left(\frac{S_t}{N}\right)I_t$ is the number of new cases, per unit time (per day), due to I_t infectious individuals.
- Example: $\beta = \frac{1}{5} = 0.2$, $\frac{S_t}{N} = \frac{20}{40} = \frac{1}{2}$ and $I_t = 10$ million. Then:
  - $\beta\left(\frac{S_t}{N}\right) = \frac{1}{5}\frac{1}{2} = \frac{1}{10}$ means that, on average, an infectious individual transmits the disease to one susceptible person (that becomes infectious) every ten days.
  - $\beta\left(\frac{S_t}{N}\right)I_t = \frac{1}{10}10 = 1$ means that, on average, one million people become infectious each day (and stop being susceptible).

Basic SIR Model

- Interpretation of γI_t.
  - γ is the average number of infectious people that become removed (through recovery or death), per unit time (per day).
  - γI_t is the number of infectious people that become removed (by recovery or death), per unit time (per day).
- Supose $\gamma = \frac{1}{10} = 0.1$ and $I_t = 10$ million. Then:
  - $\gamma = \frac{1}{10}$ means that, on average, one infectious person becomes removed every ten days. In other words: on average, the infectious period lasts ten days $(1/\gamma)$.
  - $\gamma I_t = \frac{1}{10} 10 = 1$ means that, on average, a million of the infectious become removed each day.

Basic SIR Model

- Continuous time:

$$\begin{aligned}\dot{S}(t) &= -\beta\left(\frac{I(t)}{N}\right)S(t)\\ \dot{I}(t) &= \beta\left(\frac{I(t)}{N}\right)S(t) - \gamma I(t)\\ \dot{R}(t) &= \gamma I(t)\end{aligned}$$

with $S(t) + I(t) + R(t) = N$, $S(0) = (1-\epsilon)N \cong N$, $I(0) = \epsilon N > 0$ and $R(0) = 0$.

  - For any variable X we define $\dot{X}(t) \equiv \frac{\mathrm{d}X}{\mathrm{d}t}(t)$.

- Interpretation
  - γ is the removal rate per unit of time; hence, during a short time interval h, a fraction γh of the infectious becomes removed.
    - When h is small: $R(t+h) - R(t) \cong \dot{R}(t)h = \gamma h I(t)$.
  - β is the contact rate per unit of time; hence, during a short time interval h, an individual has βh contacts (sufficient for transmission).
    - When h is small: $S(t+h) - S(t) \cong \dot{S}(t)h = -\beta h\left(\frac{I(t)}{N}\right)S(t)$.

Basic SIR Model

- We can simplify notation by dropping the time index:

$$
\begin{aligned}
\overset{\bullet}{S} &= -\beta\left(\frac{I}{N}\right)S \\
\overset{\bullet}{I} &= \beta\left(\frac{I}{N}\right)S - \gamma I \\
\overset{\bullet}{R} &= \gamma I
\end{aligned}
$$

  - We could simplify notation further by normalizing population to one: $N = 1$. In this case, S, I and R can be interpreted as fractions of the population (including the deceased). Since at $t = 0$ there are no deceased, S, I and R can be interpreted as fractions of the initial population (of living individuals).

- From now on we'll work in continuous time, but everything can be done in discrete time.

Basic SIR Model

- From $\overset{\bullet}{I}(t) = \beta \left(\frac{I(t)}{N} \right) S(t) - \gamma I(t)$ we can compute the growth rate of infectious cases at t :

$$\frac{\overset{\bullet}{I}(t)}{I(t)} = \beta \frac{S(t)}{N} - \gamma$$

- This growth rate will be positive (increasing number of infectious) when $\beta \frac{S(t)}{N} > \gamma$, that is, when

$$\mathcal{R}_t^e \equiv \frac{\beta}{\gamma} \frac{S(t)}{N} > 1$$

where $\mathcal{R}_t^e$ is the so-called **effective reproduction number** at t.

- For an epidemic to start we need a positive growth rate at $t = 0$. That is:

$$\mathcal{R}_0^e \equiv \frac{\beta}{\gamma} \frac{S(0)}{N} > 1$$

Basic SIR Model

- Since $\frac{S(0)}{N} \cong 1$, we have $\mathcal{R}_0^e \cong \frac{\beta}{\gamma}$. Hence, it is standard to define the so-called **basic reproduction number** as follows:

$$\mathcal{R}_0 \equiv \frac{\beta}{\gamma}$$

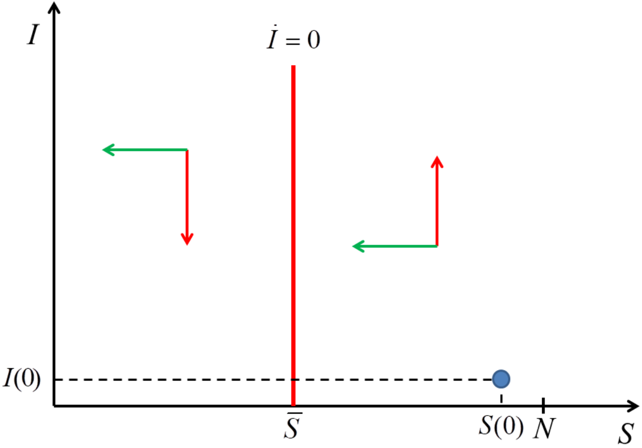
- Interpretation:
  - β is the average number of contacts of an (infectious) individual, per unit time.
  - $1/\gamma$ is the average duration of the infectious period.
  - $\mathcal{R}_0 = \beta \times \frac{1}{\gamma}$ is the expected number of new infectious individuals per sick person.
  - Example: if an individual has one contact (with transmission) every five days ($\beta = \frac{1}{5}$) and is infectious during 10 days ($\gamma = 1/10$), it will transmit the disease to $\beta \times \frac{1}{\gamma} = \frac{1}{5} \times 10 = 2$ individuals (that will become infectious).
- Remark: $\mathcal{R}_0^e \equiv \frac{\beta}{\gamma}\frac{S(0)}{N}$ and $\mathcal{R}_0 \equiv \frac{\beta}{\gamma} \Rightarrow \mathcal{R}_0^e = \mathcal{R}_0\frac{S(0)}{N}$.

- Phase Diagram
  - From $\dot{I} = \beta\left(\frac{I}{N}\right)S - \gamma I$ we can get the $\dot{I} = 0$ schedule: $\dot{I} = 0 \Rightarrow \beta\left(\frac{I}{N}\right)S - \gamma I = 0 \Rightarrow \beta\frac{S}{N} = \gamma$ (for $I > 0$). Then:

$$\begin{aligned}\overline{S} &= \frac{\gamma}{\beta}N \\ &= \frac{N}{\mathcal{R}_0} \\ &= \frac{S(0)}{\mathcal{R}_0^e}\end{aligned}$$

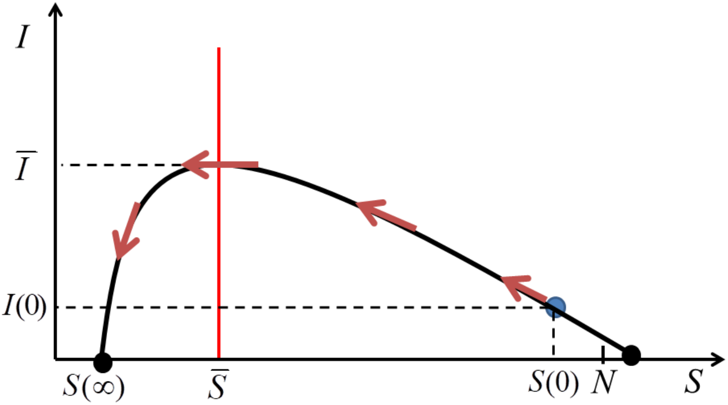
  - If $S > \overline{S}$ we get $\dot{I} > 0$, so the number of infectious is increasing over time. The opposite happens if $S < \overline{S}$.
  - From $\dot{S} = -\beta\left(\frac{I}{N}\right)S$ we see that S is always falling (as long as S and I are positive).
  - Finally, recall the initial conditions: $S(0) = (1-\epsilon)N \cong N$, $I(0) = \epsilon N > 0$.

- Phase Diagram (cont.)
  - Assuming $S(0) > \overline{S} = \frac{S(0)}{\mathcal{R}_0^e}$ (*i.e.*, $\mathcal{R}_0^e > 1$):



Basic SIR Model

- Phase Diagram (cont.)

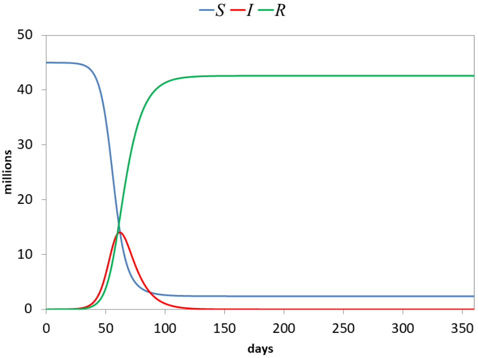


Basic SIR Model

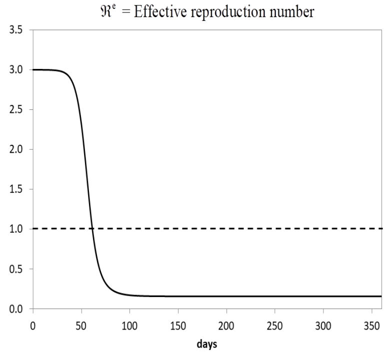
- Remarks on the previous figure
  - The equation of the black curve can be derived in closed form (see the Appendix).
  - $\overline{S}$ is the **herd immunity** threshold.
    - Once $S < \overline{S}$, the epidemic starts to disappear: I starts falling from its maximum $\overline{I}$ towards zero.
    - It can be shown that $\overline{I} = N - \overline{S} + \overline{S} \ln \left(\frac{S(0)}{\overline{S}} \right)$ (see the Appendix).
  - $S(\infty)$ denotes the long-run number of susceptibles. It is implicitly defined by $S(\infty) - \overline{S} \ln S(\infty) = N - \overline{S} \ln S(0)$ (see the Appendix).
  - $I(\infty) = 0$: no infectious individuals in the long run.
  - $N - S(\infty) - I(\infty) = N - S(\infty) = R(\infty)$ is the long-run number of removed individuals (either recovered or dead).
  - Notice that $R(\infty) = N - S(\infty)$ gives the number of persons that were sick at some time during the epidemic.

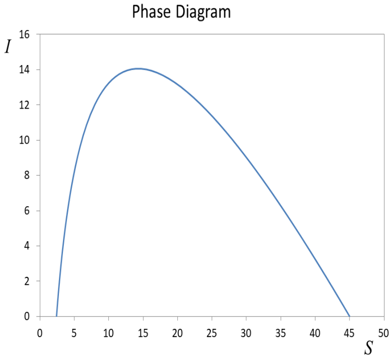
Basic SIR Model

- Given a set of parameter values, we can simulate the evolution of S, I and R.
  - We show simulations for the discrete-time version of the model, than can be easily obtained with a standard spreadsheet (*e.g.*, Excel).
- Suppose: $\gamma = 1/10$, $\beta = 0.3$, $\mathcal{R}_0 = \frac{\beta}{\gamma} = 3$, $N = 45{,}000{,}000$, $I_0 = 1{,}000$, $R_0 = 0$, $S_0 = N - I_0$.



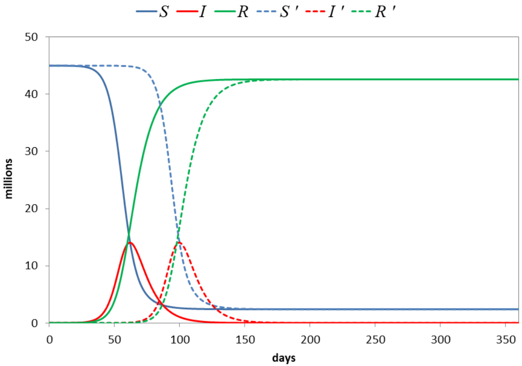
Basic SIR Model





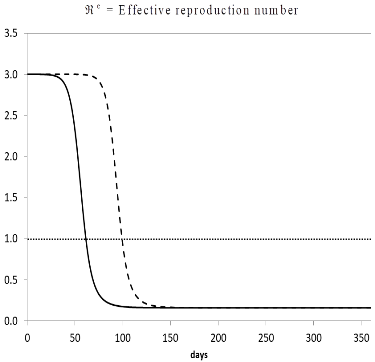
Basic SIR Model

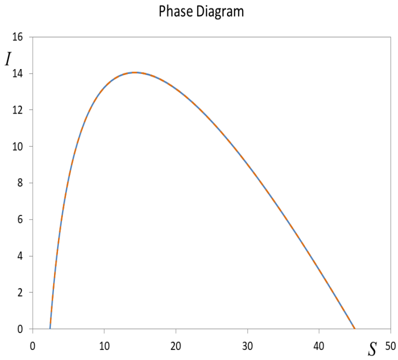
- Reduce I_0 from $1,000$ to 1, keeping the other parameters constant.



The solid lines (dashed lines) show the results for the original (new) parametrization.

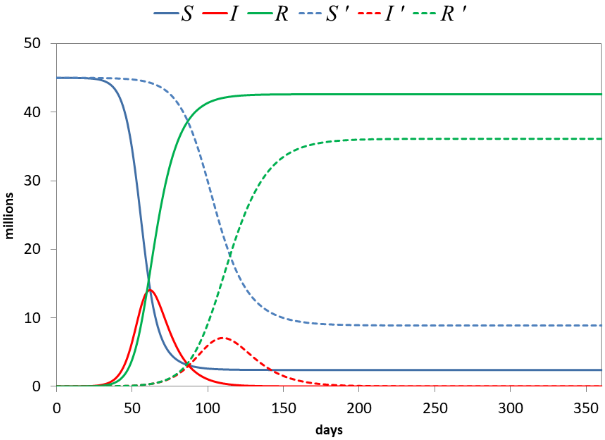
Basic SIR Model



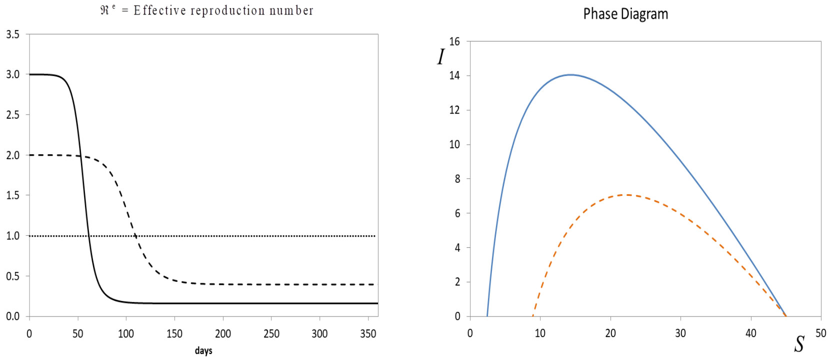


Basic SIR Model

- Reduce β from 0.3 to 0.2, keeping the other parameters constant. This reduces $\mathcal{R}_0$ from 3 to 2.

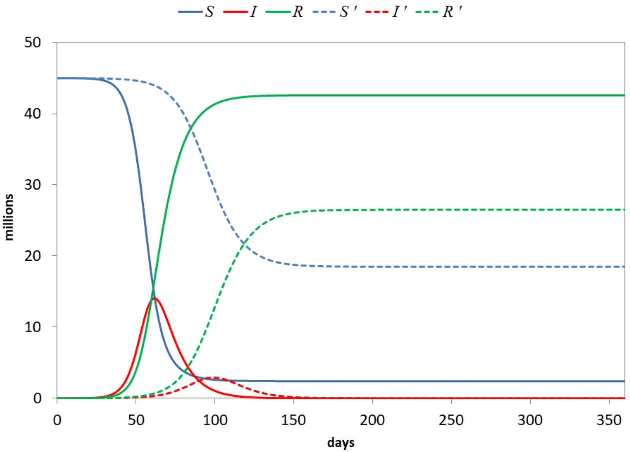


Basic SIR Model

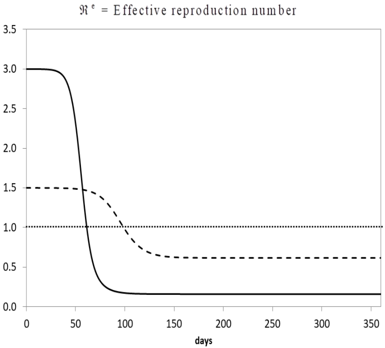


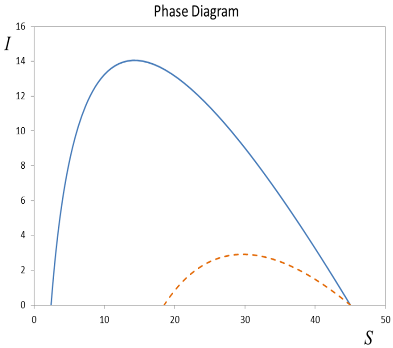
Basic SIR Model

- Increase γ from $\frac{1}{10}$ to $\frac{1}{5}$, keeping the other parameters constant. This reduces $\mathcal{R}_0$ from 3 to 1.5.



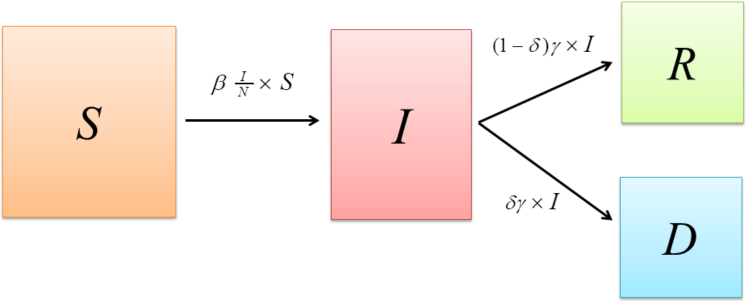
Basic SIR Model





SIRD Model

- Split the Removed into Recovered or Dead.
  - $R =$ recovered
  - $D =$ dead
  - $R + D =$ removed
- Graphically



SIRD Model

- Total population, including the deceased, is constant:

$$N = S(t) + I(t) + R(t) + D(t)$$

- We use $P(t)$ to denote the number of persons that are alive at t :

$$P(t) = S(t) + I(t) + R(t)$$

- The laws of motion are the following (dropping the time index):

$$\begin{aligned} \overset{\bullet}{S} &= -\beta\left(\frac{I}{N}\right)S \\ \overset{\bullet}{I} &= \beta\left(\frac{I}{N}\right)S - \gamma I \\ \overset{\bullet}{R} &= (1-\delta)\,\gamma I \\ \overset{\bullet}{D} &= \delta\gamma I \\ \overset{\bullet}{P} &= -\delta\gamma I \end{aligned}$$

- Initial conditions:

$$S_0 = (1-\epsilon)N \cong N, \quad I_0 = \epsilon N > 0, \quad R_0 = 0, \quad D_0 = 0$$

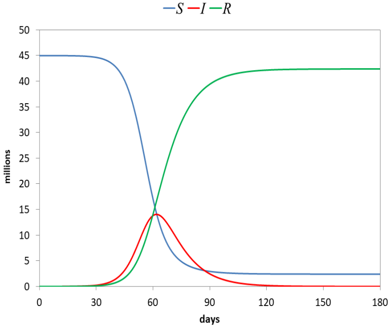
SIRD Model

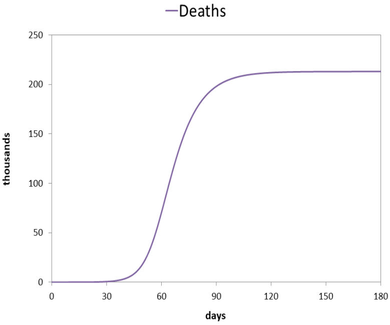
- The parameter δ is the **Case Fatality Rate** (**CFR**).
  - From $\dot{R} + \dot{D} = \gamma I$ we get $(\dot{R} + \dot{D})h = \gamma h I$. Then, in a short time interval, h, approximately $\gamma h \times I$ infectious individuals become removed. A fraction δ of those die, $\dot{D}h = \delta \gamma h I$, and a fraction $1 - \delta$ recover, $\dot{R}h = (1 - \delta)\,\gamma h I$.
  - Combining $\dot{D} = \delta \gamma I$ with $\dot{R} = (1 - \delta)\,\gamma I$ we get $\dot{D} = \frac{\delta}{1-\delta}\dot{R}$. Integrating from 0 to t, and imposing $D(0) = R(0) = 0$, we get $D(t) = \frac{\delta}{1-\delta} R(t)$. Then: $\delta = \frac{D(t)}{D(t)+R(t)}$.
  - We can think of $\delta\gamma$ as the case fatality rate per unit of time (per day). If $\delta = 0.01$ (1%) and $\gamma = \frac{1}{10}$, $\delta\gamma = 0.001$ means that 0.1% of the infectious die each day.
  - Integrating $\dot{D} = \delta \gamma I$ from 0 to t, and imposing $D(0) = 0$ we get $D(t) = \delta\gamma \int_0^t I(x)dx$. Then: $\delta\gamma = \frac{D(t)}{\int_0^t I(x)dx}$.
  - The CFR is not the same as the **mortality rate**, which is usually defined as the flow of deaths over a period of time divided by the total population (at the beginning of the period). If we take $t = 0$ as the initial time we get: $mortality\ rate_t = \frac{D(t)-D(0)}{P(0)} = \frac{D(t)}{N}$.

SIRD Model

- In reality, the CFR may not be constant. One possibility is that there are congestion effects in the health care system. This can be modeled by assuming that the CFR is an increasing function of the number of infectious people (or a fraction of the infectious people, reflecting those that may need hospitalization).
  - Examples used in the literature: $\delta(t) = \varphi + \kappa I(t)$; $\delta(t) = \varphi + \kappa I(t)^2$.
- In the SIRD model, the effective reproduction number ($\mathcal{R}_t^e$) and the basic reproduction number ($\mathcal{R}_0$) are the same as those in the basic SIR model.
- The structure of the phase diagram in (S, I) space is also the same.
- The dynamics for S, I, and the number of removed individuals are exactly the same as those induced by the basic SIR model. But now we have the extra information about the number of dead people (D).
- In the equations for $\dot{S}$ and $\dot{I}$, I is divided by N (which includes the deceased). This is standard in the literature, but it would be better to divide by P. Usually this is not problematic, since the number of deaths is relatively small. As mentioned earlier, N is sometimes normalized to 1, and variables are interpreted as fractions of the initial population (of living individuals).

- Suppose, again, that $\gamma = 1/10$, $\beta = 0.3$, $\mathcal{R}_0 = \frac{\beta}{\gamma} = 3$, $N = 45,000,000$ and $I_0 = 1,000$. Also: $R_0 = D_0 = 0$ and $S_0 = N - I_0$. Assume $\delta = 0.005$.

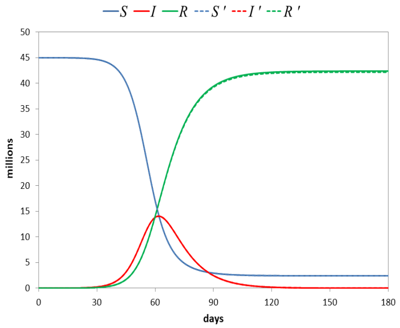


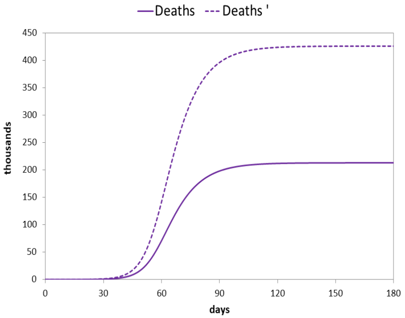


- Even with a small fatality rate (0.5%) the number of deaths is high ($213,000$ in a population of 45 million).

SIRD Model

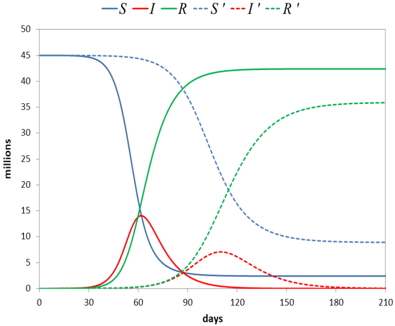
- A higher fatality rate increases the number of deaths and reduces the number of recovered, without affecting the number of removed individuals. Also, S and I remain unaffected.
- $\delta = 0.01$ (1%)

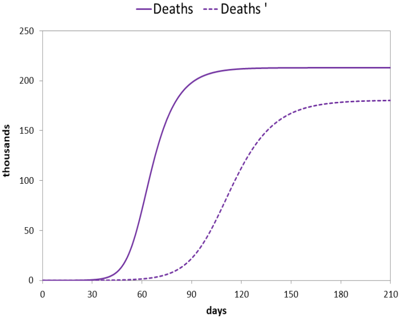




SIRD Model

- Reduce β from 0.3 to 0.2, keeping the other parameters constant. This reduces $\mathcal{R}_0$ from 3 to 2.





- The number of deaths falls from $213,000$ to $180,500$.

SIRD Model with Lockdown

- Suppose the government responds to the epidemic by mandating a uniform lockdown.
- Let $L(t)$ denote the fraction of $P(t)$ that is in lockdown at t.
- Assume $L(t) \in \left[0, \overline{L}\right]$, with $\overline{L} < 1$.
  - Full lockdown is not feasible: some goods are essential, etc.
- Let $\theta(t) \in [0, 1]$ denote the effectiveness of the lockdown at t.
  - $\theta = \frac{3}{4}$ means that 25% of the individuals do not obey the lockdown.
- The effective lockdown at t is $\theta(t)L(t)$.
- The number of susceptibles that can become infectious is now $[1 - \theta(t)L(t)]S(t)$.
- The number of infectious that can transmit the disease is now $[1 - \theta(t)L(t)]I(t)$.

SIRD Model with Lockdown

- Normalizing $N = 1$, and dropping the time index, we get the following equations:

$$\begin{aligned}
\overset{\bullet}{S} &= -\beta\left(1-\theta L\right) I \left(1-\theta L\right) S \\
\overset{\bullet}{I} &= \beta\left(1-\theta L\right) I \left(1-\theta L\right) S - \gamma I \\
\overset{\bullet}{R} &= \left(1-\delta\right)\gamma I \\
\overset{\bullet}{D} &= \delta\gamma I \\
\overset{\bullet}{P} &= -\delta\gamma I
\end{aligned}$$

SIRD Model with Lockdown

- We can rewrite the equations for $\overset{\bullet}{S}$ and $\overset{\bullet}{I}$ as follows:

$$\overset{\bullet}{S} = -\beta\left(1-\theta L\right)^2 IS$$

$$\overset{\bullet}{I} = \beta\left(1-\theta L\right)^2 IS - \gamma I$$

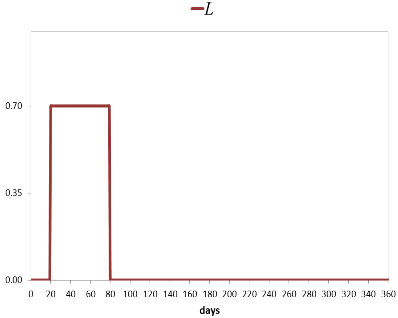
- Notice that the lockdown works like a reduction in the contact rate.
  - If we define the effective contact rate as $\beta^e(t) \equiv \beta[1-\theta(t)L(t)]^2$ we get (dropping the time index): $\overset{\bullet}{S} = -\beta^e IS$ and $\overset{\bullet}{I} = \beta^e IS - \gamma I$.
- The effective reproduction number is now (recall we have normalized $N = 1$):

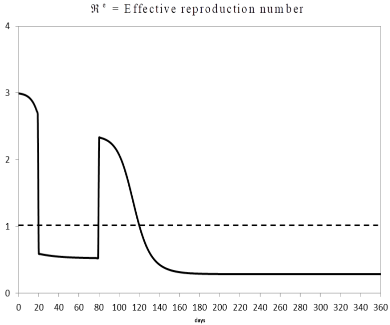
$$\mathcal{R}_t^e \equiv \frac{\beta}{\gamma}[1-\theta(t)L(t)]^2 S(t) = \frac{\beta^e(t)}{\gamma} S(t)$$

- The basic reproduction number is always calculated in the absence of any policy. Hence, it is still given by $\mathcal{R}_0 \equiv \frac{\beta}{\gamma}$.

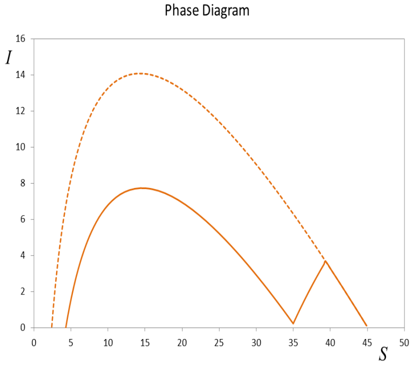
SIRD Model with Lockdown

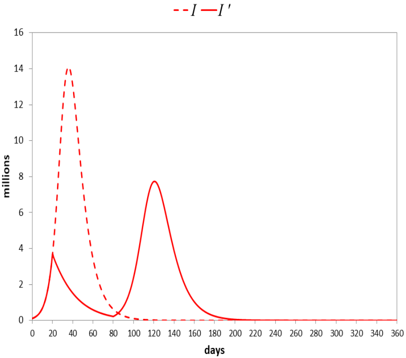
- Suppose $\gamma = 1/10$, $\delta = 0.005$, $\beta = 0.3$, $\mathcal{R}_0 = \frac{\beta}{\gamma} = 3$, $N = 45,000,000$ and $I_0 = 112,500$, $R_0 = D_0 = 0$ and $S_0 = N - I_0$.
- In addition, assume $L(t) = 0.7 \; \forall \in [20, 80)$ and $L(t) = 0$ otherwise, with constant effectiveness $\theta = 0.75$.





SIRD Model with Lockdown

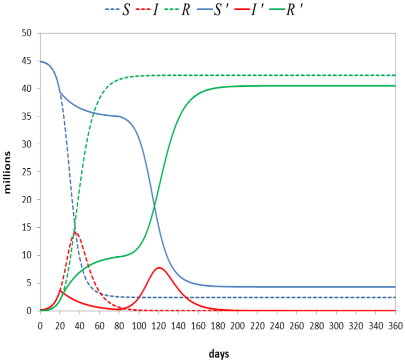


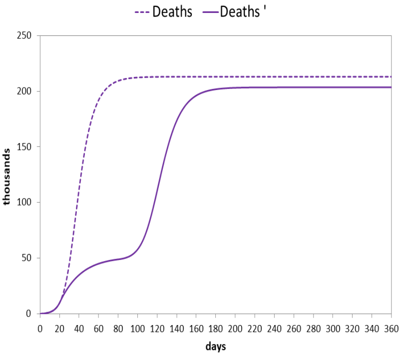


For comparison, the dashed lines show the case with no lockdown.

Notice the emergence of a second wave of infections when the lockdown is lifted.

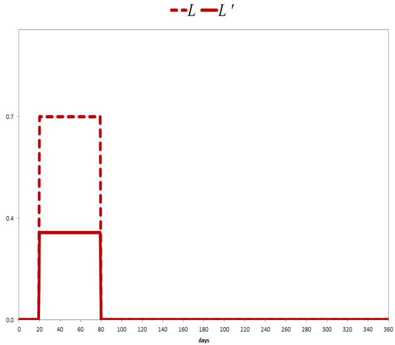
SIRD Model with Lockdown

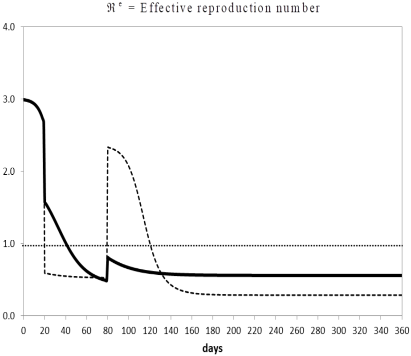




Notice that the final number of deaths is somewhat smaller but still high.

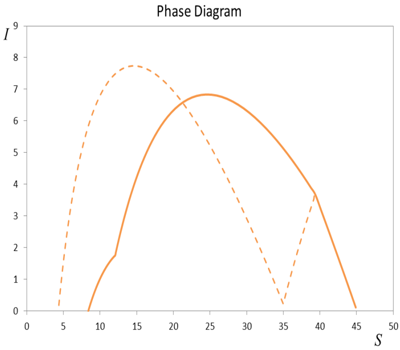
- A milder lockdown

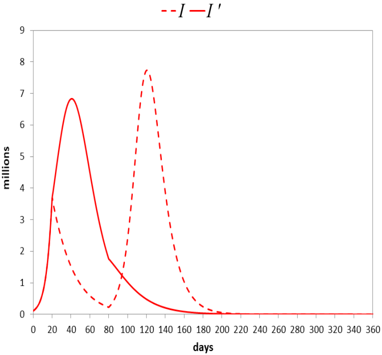




For comparison, the dashed lines show the case with a tighter lockdown.

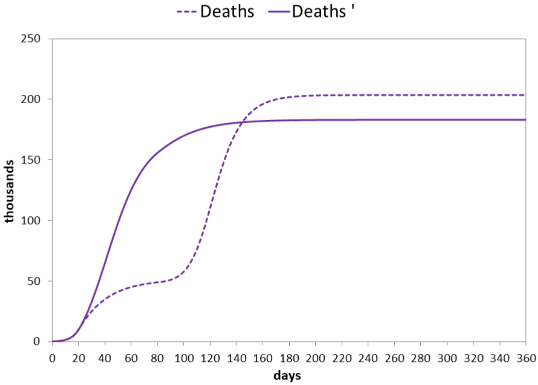
SIRD Model with Lockdown





No second wave with the milder lockdown. Herd immunity is reached earlier.

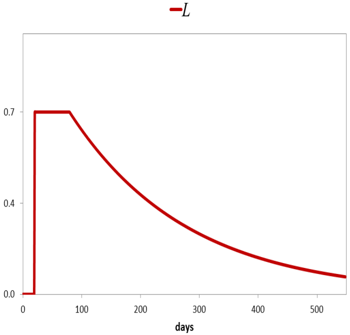
SIRD Model with Lockdown

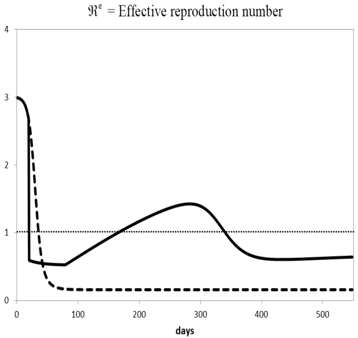


Notice that, in the end, the total number of deaths is smaller with the milder lockdown.

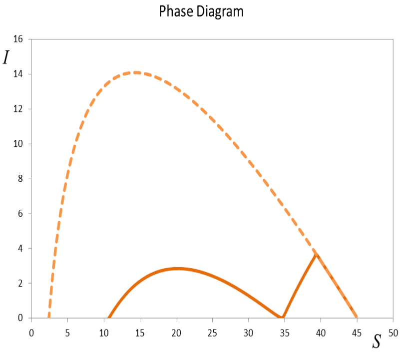
SIRD Model with Lockdown

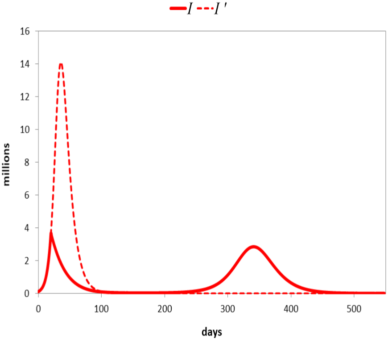
- A tight lockdown that is lifted gradually.
  - The rest of the parameters remain unchanged.



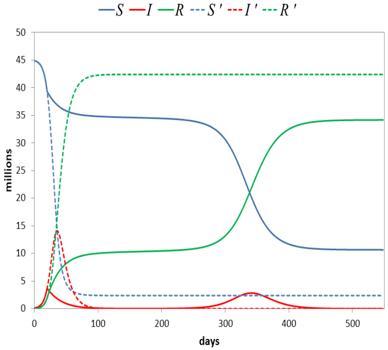


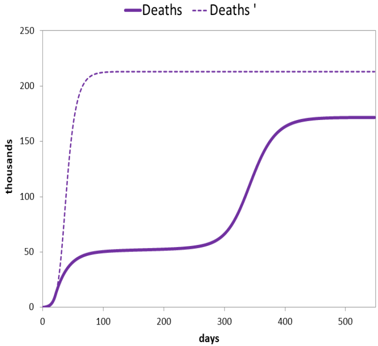
For comparison, the dashed lines show the case with no lockdown.


Phase Diagram
I
S
16
14
12
10
8
6
4
2
0
0
5
10
15
20
25
30
35
40
45
50


I
I '
millions
days
16
14
12
10
8
6
4
2
0
0
100
200
300
400
500

SIRD Model with Lockdown





SIRD Model: Testing and Tracing

- Consider the SIRD model with lockdown.
- We want to introduce a testing-tracing-quarantine (TTQ) policy.
- Suppose that, at time t, the government can identify (trace), test as infectious, and subsequently quarantine, a fraction $\psi(t)$ of $I(t)$.
- Assume quarantined individuals don't transmit the virus. Hence, the relevant number of infectious in the laws of motion for $S(t)$ and $I(t)$ is $[1-\psi(t)]I(t)$.
- Then, dropping the time index and normalizing $N=1$, we get:

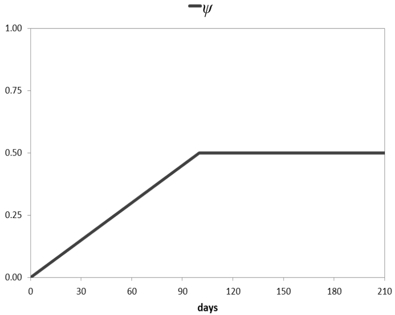
$$\begin{aligned} \dot{S} &= -\beta\left(1-\theta L\right)^{2}(1-\psi)IS \\ \dot{I} &= \beta\left(1-\theta L\right)^{2}(1-\psi)IS-\gamma I \end{aligned}$$

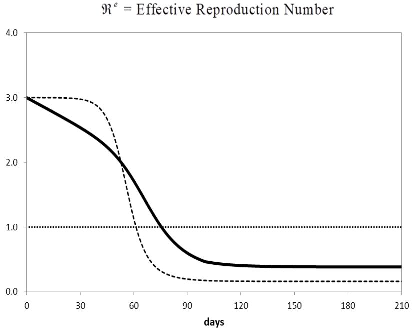
- The rest of the equations remain unchanged.
- The effective reproduction number is now

$$\begin{aligned} \mathcal{R}_t^e &\equiv \frac{\beta}{\gamma}[1-\theta(t)L(t)]^2[1-\psi(t)]S(t) \\ &= \frac{\beta^e(t)}{\gamma}S(t) \end{aligned}$$

SIRD Model: Testing and Tracing

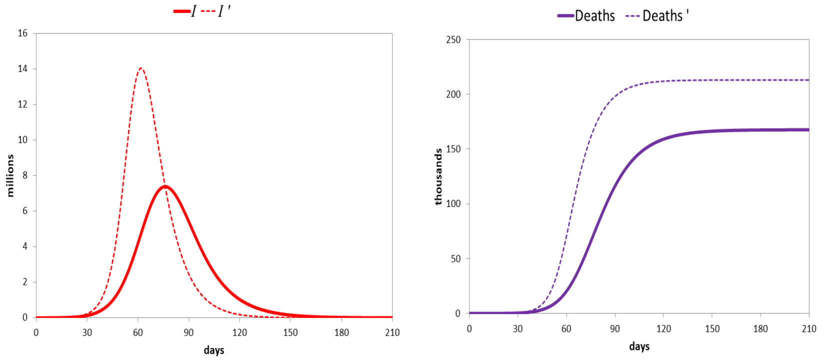
- Suppose $\gamma = 1/10$, $\delta = 0.005$, $N = 45,000,000$ and $I_0 = 1000$, $R_0 = D_0 = 0$ and $S_0 = N - I_0$.
- Let $\psi(t)$ increase linearly from 0 to 0.5 during the first one hundred days and remain there afterwards. There's no lockdown.





The dashed lines show the case of no testing.

SIRD Model: Testing and Tracing



SIRD Model: Social Distancing in reduced form

- We've seen that, in the model with lockdown and TTQ, the equations for $\dot{S}$ and $\dot{I}$ can be rewritten as $\dot{S}(t) = -\beta^e(t)I(t)S(t)$ and $\dot{I}(t) = \beta^e(t)I(t)S(t) - \gamma I(t)$, with $\beta^e(t) \equiv \beta[1-\theta(t)L(t)]^2[1-\psi(t)]$.
- This suggests that we could model social-distancing measures, including the endogenous response of individuals, through a time-varying contact rate, $\beta(t)$.
- In this case, the equations for $\dot{S}$ and $\dot{I}$ become

$$
\begin{aligned}
\dot{S}(t) &= -\beta(t)I(t)S(t) \\
\dot{I}(t) &= \beta(t)I(t)S(t) - \gamma I(t)
\end{aligned}
$$

and we need a description of the evolution of $\beta(t)$ over time.

  - The rest of the equations of the model remain unaffected.

SIRD Model: Social Distancing in reduced form

- Suppose

$$\beta(t) = \beta_0 e^{-\lambda t} + \beta_*(1 - e^{-\lambda t})$$

  where $\beta_0 > 0$, $\beta_* > 0$ and $\lambda > 0$.

- With this functional form, $\beta(t)$ starts at β_0 and converges monotonically to β_* at a rate determined by λ.
  - The standard case with $\beta(t)$ constant is obtained by setting $\beta_0 = \beta_*$.

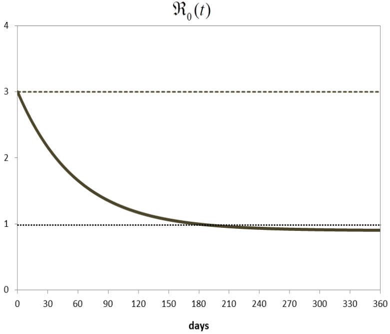
- If we define $\mathcal{R}_0(t) \equiv \frac{\beta(t)}{\gamma}$, the expression above is equivalent to

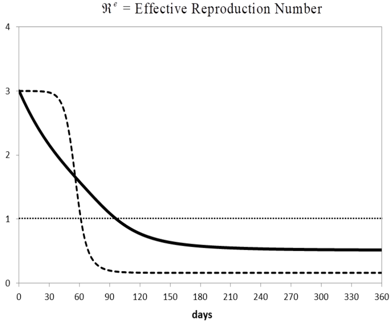
$$\mathcal{R}_0(t) = \mathcal{R}_0 e^{-\lambda t} + \mathcal{R}_*(1 - e^{-\lambda t})$$

  where $\mathcal{R}_0 \equiv \frac{\beta_0}{\gamma}$ and $\mathcal{R}_* \equiv \frac{\beta_*}{\gamma}$.

SIRD Model: Social Distancing in reduced form

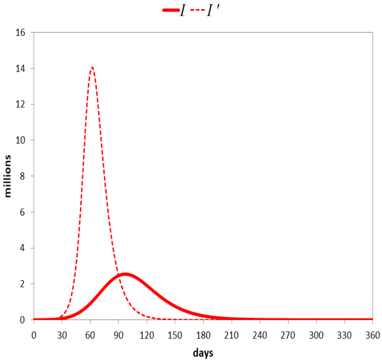
- Suppose $\gamma = 1/10$, $\delta = 0.005$, $N = 45,000,000$ and $I_0 = 1000$, $R_0 = D_0 = 0$ and $S_0 = N - I_0$.
- Also, set $\beta_0 = 0.3$ ($\mathcal{R}_0 = 3$), $\beta_* = 0.09$ ($\mathcal{R}_* = 0.9$), and $\lambda = 0.017$.

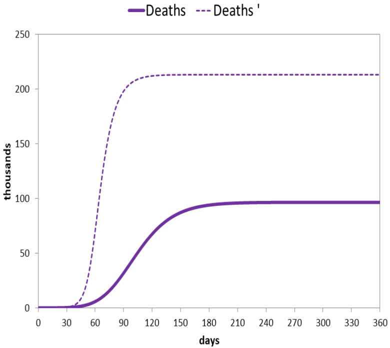




The dashed lines show the case of constant β.

SIRD Model: Social Distancing in reduced form





SIRD Model: Vaccine

- Consider a standard SIRD model with lockdown.
- Suppose a vaccine is discovered at $t = \tau$.
- Assume, for simplicity, that all susceptible individuals can be instantaneously vaccinated at $t = \tau$.
  - Suppose infectious individuals are all symptomatic and can be identified as such. In this case, there's no need to vaccinate them. Recovered individuals need not be vaccinated either.
- Before τ, the variables evolve according to the equations given earlier for the SIRD model with lockdown.
- At $t = \tau$ all susceptibles become recovered. Then:

$$\begin{aligned} S(\tau) &= 0 \\ R(\tau) &= R\left(\tau^{-}\right) + S\left(\tau^{-}\right) \end{aligned}$$

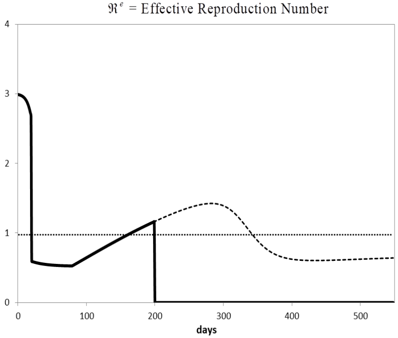
SIRD Model: Vaccine

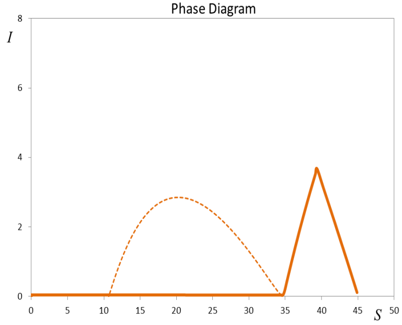
- For $t > \tau$ we have:

$$
\begin{aligned}
S(t) &= 0 \qquad\qquad (\Rightarrow \overset{\bullet}{S}(t) = 0) \\
\overset{\bullet}{I}(t) &= -\gamma I(t) \\
\overset{\bullet}{R}(t) &= (1-\delta)\,\gamma I(t) \\
\overset{\bullet}{D}(t) &= \delta\gamma I(t) \\
\overset{\bullet}{P}(t) &= -\delta\gamma I(t)
\end{aligned}
$$

SIRD Model: Vaccine

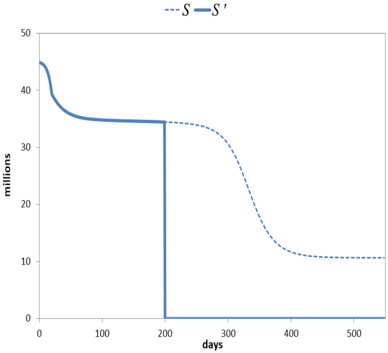
- Consider our previous simulation with a lockdown that is lifted gradually.
- Assume a vaccine becomes available at $t = 200$.

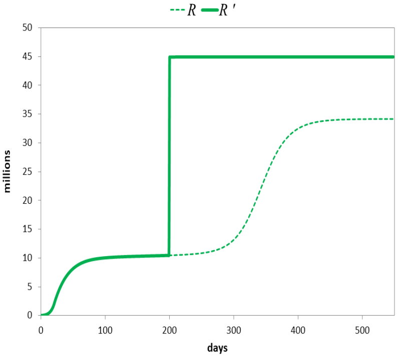




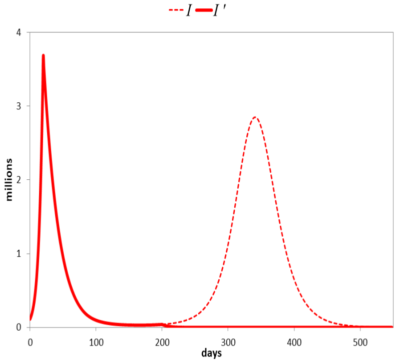
The dashed lines show the case without the arrival of the vaccine.

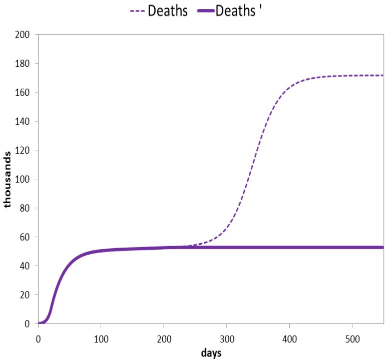
SIRD Model: Vaccine





SIRD Model: Vaccine



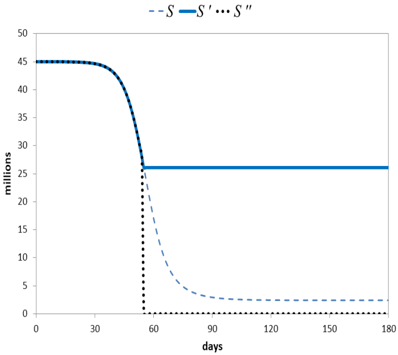


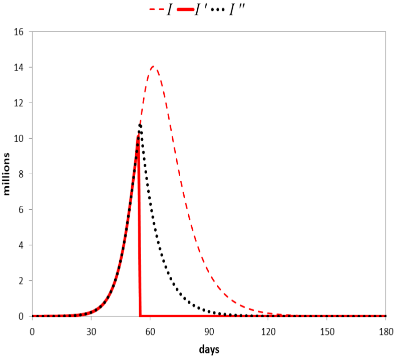
SIRD Model: Treatment

- Consider a standard SIRD model.
- Suppose a treatment that cures the disease becomes available at $t = \tau$.
- Before τ, the variables evolve according to the equations given earlier for the SIRD model with lockdown.
- Assume that, once it becomes available, the treatment is provided to all infectious individuals at every $t \geq \tau$, who are instantly transformed into recovered individuals.
- As a result, the number of new deaths from the disease goes to zero (the stock of deceased stabilizes).
- Since there are no more infectious individuals, the number of susceptibles stabilizes.
- Then:

$$
\begin{aligned}
I(t) &= 0 \quad \forall t \geq \tau \\
S(t) &= S\left(\tau^{-}\right) \quad \forall t \geq \tau \\
D(t) &= D\left(\tau^{-}\right) \quad \forall t \geq \tau \\
R(t) &= N - D\left(\tau^{-}\right) - S\left(\tau^{-}\right) \quad \forall t \geq \tau
\end{aligned}
$$

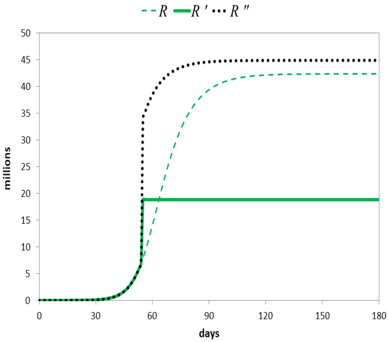
SIRD Model: Treatment

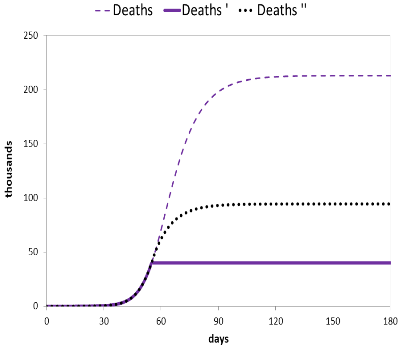




Dashed lines show the case with no treatment and no vaccine; dotted lines show the case with a vaccine.

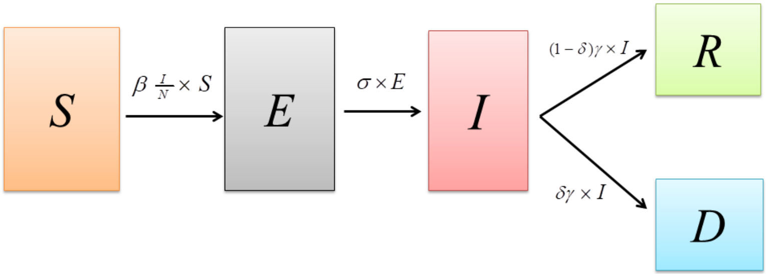
SIRD Model: Treatment





Extensions: SEIRD Model

- New state: E = *exposed*
  - Individuals that are infected but not yet infectious (incubation period).
- $N = S + E + I + R + D$ (constant).
- $P = S + E + I + R$.



Extensions: SEIRD Model

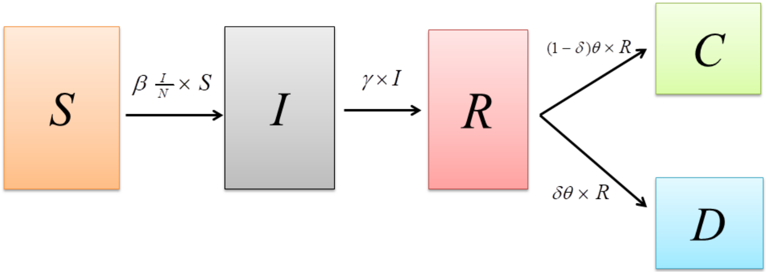
- Normalizing $N = 1$:

$$
\begin{aligned}
\overset{\bullet}{S} &= -\beta I S \\
\overset{\bullet}{E} &= \beta I S - \sigma E \\
\overset{\bullet}{I} &= \sigma E - \gamma I \\
\overset{\bullet}{R} &= (1-\delta)\,\gamma I \\
\overset{\bullet}{D} &= \delta \gamma I \\
\overset{\bullet}{P} &= -\delta \gamma I
\end{aligned}
$$

- $\sigma = \frac{1}{5}$ means that, on average, the incubation period lasts five days.

Extensions: SIRCD Model

- Now $R =$ *resolving*, and $C =$ *reCovered*.
- $N = S + I + R + C + D$ (constant).
- $P = S + I + R + C$.



Extensions: SIRCD Model

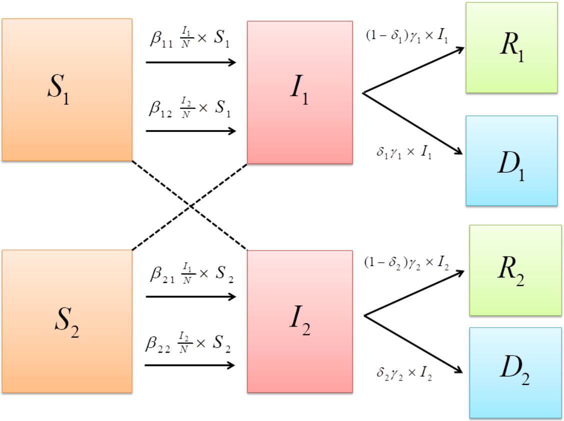
- Normalizing $N = 1$:

$$
\begin{aligned}
\dot{S} &= -\beta I S \\
\dot{I} &= \beta I S - \gamma I \\
\dot{R} &= \gamma I - \theta R \\
\dot{C} &= (1 - \delta)\, \theta R \\
\dot{D} &= \delta \theta R \\
\dot{P} &= -\delta \theta R
\end{aligned}
$$

- Now $\gamma = \frac{1}{5}$ means that, on average, a person with the virus is infectious for five days, while $\theta = \frac{1}{10}$ means that, on average, an infected person that stopped being infectious will be ill for another 10 days.

Extensions: Multi-group SIRD Model with Lockdown

- Two groups (for simplicity): $g \in \{1, 2\}$.
- Useful to model heterogeneity: different fatality rates by age, targeted lockdowns, etc.
- Figure (without lockdown):



Extensions: Multi-group SIRD Model with Lockdown

- We have:

$$
\begin{aligned}
N_g &= S_g + I_g + R_g + D_g \quad \text{(constant)} \\
P_g &= S_g + I_g + R_g \\
N &= N_1 + N_2 \\
P &= P_1 + P_2 \\
S &= S_1 + S_2 \\
I &= I_1 + I_2 \\
R &= R_1 + R_2 \\
D &= D_1 + D_2
\end{aligned}
$$

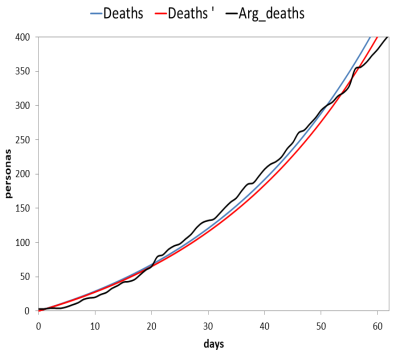
Extensions: Multi-group SIRD Model with Lockdown

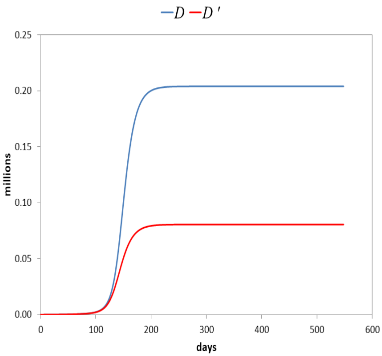
- Laws of motion, normalizing $N = 1$:

$$
\begin{aligned}
\dot{S}_1 &= -\beta_{11}\left(1-\theta_1 L_1\right) I_1\left(1-\theta_1 L_1\right) S_1-\beta_{12}\left(1-\theta_2 L_2\right) I_2\left(1-\theta_1 L_1\right) S_1 \\
\dot{S}_2 &= -\beta_{21}\left(1-\theta_1 L_1\right) I_1\left(1-\theta_2 L_2\right) S_2-\beta_{22}\left(1-\theta_2 L_2\right) I_2\left(1-\theta_2 L_2\right) S_2 \\
\dot{I}_1 &= \beta_{11}\left(1-\theta_1 L_1\right) I_1\left(1-\theta_1 L_1\right) S_1+\beta_{12}\left(1-\theta_2 L_2\right) I_2\left(1-\theta_1 L_1\right) S_1-\gamma_1 I_1 \\
\dot{I}_2 &= \beta_{21}\left(1-\theta_1 L_1\right) I_1\left(1-\theta_2 L_2\right) S_2+\beta_{22}\left(1-\theta_2 L_2\right) I_2\left(1-\theta_2 L_2\right) S_2-\gamma_2 I_2 \\
\dot{R}_1 &= \left(1-\delta_1\right) \gamma_1 I_1 \\
\dot{R}_2 &= \left(1-\delta_2\right) \gamma_2 I_2 \\
\dot{D}_1 &= \delta_1 \gamma_1 I_1=-\dot{P}_1 \\
\dot{D}_2 &= \delta_2 \gamma_2 I_2=-\dot{P}_2
\end{aligned}
$$

Estimation Difficulties

- Identification problem: δ and $I(0)$.





- We are going to derive a closed-form solution for the evolution of the epidemic in (S, I) space.
- From $\dot{I}(t) = \beta\left(\frac{I(t)}{N}\right) S(t) - \gamma I(t)$ and $\dot{S}(t) = -\beta\left(\frac{I(t)}{N}\right) S(t)$ we get:

$$\frac{\dot{I}(t)}{\dot{S}(t)} = \frac{\beta\left(\frac{I(t)}{N}\right) S(t) - \gamma I(t)}{-\beta\left(\frac{I(t)}{N}\right) S(t)}$$

$$\frac{\dot{I}(t)}{\dot{S}(t)} = -1 + \frac{\gamma}{\beta}\frac{N}{S(t)}$$

- Then:

$$\dot{I}(t) = -\dot{S}(t) + \overline{S}\frac{\dot{S}(t)}{S(t)}$$

Appendix

- Integrating both sides of the previous expression from 0 to t :

$$\int_0^t \overset{\bullet}{I}(x)\,dx = -\int_0^t \overset{\bullet}{S}(x)\,dx + \overline{S}\int\limits_0^t \frac{\overset{\bullet}{S}(x)}{S(x)}dx$$

- Then:

$$\begin{aligned} I(x)]_0^t &= -[S(x)]_0^t + \overline{S}[\ln S(x)]_0^t \\ \\ I(t) - I(0) &= -[S(t) - S(0)] + \overline{S}[\ln S(t) - \ln S(0)] \end{aligned}$$

- Then:

$$I(t) = \left[I(0) + S(0) - \overline{S}\ln S(0)\right] + \overline{S}\ln S(t) - S(t)$$

- Since $I(0) + S(0) = N$:

$$I(t) = \left[N - \overline{S}\ln S(0)\right] + \overline{S}\ln S(t) - S(t)$$

Appendix

- Dropping the time index from the previous expression we get:

$$I = \left[N - \overline{S}\ln S\left(0\right)\right] + \overline{S}\ln S - S$$

which gives the equation for the black line in the phase diagram.

- Now we can solve for the value of S that maximizes de number of infectious, I.
- Differentiating the expression above with respect to S and equating it to zero we get: $\overline{S}\frac{1}{S} - 1 = 0$. Then:

$$S = \overline{S}$$

which coincides with the value given in the phase diagram.

- Evaluating the objective function at $S = \overline{S}$ we get: $\overline{I} = N - \overline{S}\ln S\left(0\right) + \overline{S}\ln \overline{S} - \overline{S} \Rightarrow \overline{I} = N - \overline{S} - \overline{S}[\ln S\left(0\right) - \ln \overline{S}] \Rightarrow$

$$\overline{I} = N - \overline{S} + \overline{S}\ln\left(\frac{S(0)}{\overline{S}}\right)$$

Appendix

- We can compute the long-run number of susceptibles, $S(\infty)$, by setting $I = 0$ in $I = \left[N - \overline{S}\ln S\left(0\right)\right] + \overline{S}\ln S - S$.
- We get: $0 = N - \overline{S}\ln S\left(0\right) + \overline{S}\ln S\left(\infty\right) - S\left(\infty\right) \Rightarrow$

$$S\left(\infty\right) - \overline{S}\ln S\left(\infty\right) = N - \overline{S}\ln S\left(0\right)$$

- The expression above implicitly defines the value of $S\left(\infty\right)$.
    - This equation has a solution with $S\left(\infty\right) > N$, which is not the one we're looking for (we can see this from the phase diagram). We are looking for a solution with $S\left(\infty\right) < \overline{S} < N$.

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